Krish Bhagirath

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Education

McMaster University **Hamilton, ON**

Bachelor of Engineering in Computer Engineering *Sep. 2024 - Apr. 2028*

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Intro to Programming, Statistics for Engineering, Data Structures & Algorithms
- Awarded $20,000+ in merit-based entrance scholarships for academic excellence.

Experience

AI/ML Engineer Intern **Apr. 2026 – Present**

MobilitiQ *Hamilton, ON*

- Accelerated Austin ride-hailing simulation **19×** using A* routing and k-d tree candidate pruning.
- Engineered a discrete-event EMS simulator with precomputed Dijkstra travel-time tables and MEXCLP ambulance allocation; simulated **6,000** calls for **300** ambulances on a 75K-node graph in **1.8s**.
- Developed TensorFlow LSTM forecasts on **757K+** Austin trips; seasonal ensembling cut offline MAE by **17%**.
- Trained PPO across **128** parallel simulators; served ONNX policies via FastAPI with **8e-7** PyTorch parity.
- Built FastAPI/React Native EMS replay, generating 300-unit map frames in **6.8ms** median latency.

Software Engineer Intern **May 2025 – Aug. 2025**

Novologic *Toronto, ON*

- Refactored **15+** Laravel controllers into dependency-injected services and repositories; integrated PHPUnit regression suites into GitHub Actions CI, increasing code coverage from **65% to 100%**.
- Reduced dashboard API p95 latency from **480 to 210 ms** via Eloquent eager loading and composite MySQL indexes.
- Deployed a React RAG assistant with Gemini and semantic document search, utilized by **20+ clients** daily.
- Automated HubSpot lead intake with webhooks and idempotent upserts, processing **250 inquiries/month**.
- Secured **750+** document uploads across **6+** clients using AWS S3 presigned URLs and tenant-scoped authorization.

Machine Learning Developer **Oct. 2025 – Present**

McMaster Aerial Robotics & Drone Club *Hamilton, ON*

- Benchmarked OpenCV HoughCircles and fine-tuned CNN detectors, yielding **95%** blob-detection accuracy within 5m.
- Developing software-in-the-loop (**SITL**) drone simulations on **Linux VM** with **DroneKit** to validate autonomy logic.
- Building a **2D LiDAR** wall-following system on **Jetson Nano** as a lightweight alternative to SLAM-based navigation.

Projects

PartnerUp ↗ | *TypeScript, Next.js, Node.js, PostgreSQL, Playwright, GCP*

- Built a Next.js/TypeScript lab-partner matching platform for McMaster students, tested by **25+** users prelaunch.
- Automated Mosaic schedule imports by deploying a **Playwright** worker on **GCP** to parse and deduplicate schedules into **PostgreSQL**, reducing manual section entry.

DeepFakeDetector ↗ | *Python, PyTorch, CNN, Vision Transformer, FFT, Next.js, FastAPI*

- Developed a **4-branch** hybrid deepfake detection model combining CNN, ViT, FFT, and gradient-based features for **cross-generator generalization**, achieving up to **96.7%** accuracy on AI-GenBench and OpenFake benchmarks.
- Enhanced ViT performance by **14%** via fine-tuning DeiT models with MLP heads and differential learning rates.
- Implemented a **Next.js and FastAPI** web app, utilizing **Grad-CAM** for visual heatmaps of deepfake predictions.

GitCheck ↗ | *LangChain, Gemini API, GitHub API, Python, FastAPI, Next.js, Supabase*

- Built a **validation agent** with LangChain and Python to eliminate manual hackathon submission review.
- Designed a **web-scraping pipeline** linking Devpost submissions to their GitHub repos for automated evaluation.
- Integrated the **GitHub API** to analyze commit patterns and repository structure to enforce authenticity checks.
- Deployed a **Next.js** frontend and **FastAPI** backend generating automated validation scores for organizer decisions.

VALID ↗ | *Python, Pandas, NumPy, scikit-learn, PyTorch, XGBoost, ClinicalBERT*

- Built a **CT-scan decision model** to predict imaging necessity at ED triage by developing a ClinicalBERT-based ensemble (MLP, Random Forest, XGBoost), achieving **91% precision** and **88% recall.**
- Preprocessed **127K+** MIMIC-IV EHR records via **Pandas/NumPy** for data cleaning and feature selection/encoding.
- Awarded the **Responsible AI Award** at CUCAI among **70+ teams** for strong clinical safety and medical impact.

Technical Skills

Programming Languages: Python, C++, C, Java, Typescript, JavaScript, PHP, SQL, SystemVerilog
Frameworks & Libraries:PyTorch, TensorFlow, Scikit-learn, React, Next.js, FastAPI, NumPy, Pandas
Tools & Technologies: Git, Linux, PostgreSQL, MySQL, Docker, GCP, AWS, Playwright